CLEMSON UNIVERSITY

CPSC - 8650 - 500
Data Mining
Spring 2026

Course Description

Data mining is now a powerful and evolving force across diverse fields such as computer science, bioinformatics, business, health sciences, medical sciences, and industrial engineering. Recent improvements in data mining techniques are a key enabler of the latest breakthroughs in artificial intelligence. By leveraging the immense volumes of data available – including consumer data, medical records, web information, and social media – data mining enables the discovery of valuable hidden insights. This significant potential has driven the rapid expansion of the data mining software market into a multi-billion-dollar sector. This graduate-level course will empower students with a comprehensive knowledge of data mining algorithm design and implementation, exposure to cutting-edge research, and practical skills through real-world applications.

Course Learning Objectives

- Understand the role of data mining in knowledge discovery.
- Be familiar with the characteristics of different data types and master various data preprocessing techniques.
- Understand various data mining functionalities and how they can be applied to various real-world problems.
- Fully comprehend traditional data mining methods and techniques, such as association rules, clustering, and classification, and know how to use them to solve real-world problems.
- Get to know the advanced data mining techniques, such as deep learning, and understand how they advance emerging applications such as AI.
- Be able to design, train, and evaluate various data mining algorithms and apply them to real-world datasets.

Course Schedule

- fundamental concepts
- data preprocessing
- information filtering
- characterization and comparison
- frequent pattern analysis
- classification
- cluster analysis
- neural networks
- deep learning
- outlier analysis
- data mining trends and research frontiers

Course Information

Course Modality:

Required Materials

Class Materials are available for purchase through the [Clemson Bookstore.]

Data Mining: Concepts and Techniques
ISBN: 978-0-12-811760-6
Authors: Jiawei Han, Micheline Kamber and Jian Pei
Publisher: Morgan Kaufmann Publishers, An imprint of Elsevier
Publication Date: July 2, 2022
Edition: 4th Edition

Course Grading Scheme and Grading Policies

Course Grading Policies:

- Homework (40%): Assignments will be given throughout the entire semester.
- Test #1 (15%): Cover the contents studied in the first quarter of the semester.
- Test #2 (15%): Cover the contents studied in the second quarter of the semester.
- Test #3 (15%): Cover the contents studied in the third quarter of the semester.
- Test #4 (15%): Cover the contents studied in the fourth quarter of the semester.

Grading Scheme:

A (90 - 100), B (80 - 89), C (70 - 79), D (60 - 69), F (0 - 59)

Course Absence/Attendance Policy

- You must finish your homework assignments independently. Any cheating will result in a zero point on that assignment for both parties.
- You must submit your homework by the due date. Late submissions will be penalized by 10% for each late day for up to three days. Submissions past the deadline by three days will get zero in grade.

Experiential Learning

This optional component provides suggested syllabi language if your course uses Experiential Learning pedagogy. If your course is not using Experiential Learning pedagogy, this component can be disabled by toggling the "visible" option.

The intention is for the text in black to remain and text in red can be adjusted to match your course and how you intend to address each component of Kolb’s Cycle. This text should be removed before publishing your syllabus.

This course provides a Clemson experiential learning (CU-ExL) opportunity which follows Kolb’s experiential learning cycle (1984). Research shows that Kolb's model of learning positively impacts student learning through facilitated, hands-on experiences which encourage students to connect theories and knowledge to real-world situations

***CU-ExL Student Learning Outcome**: Students will engage in academic learning through real-world, concrete experiences supporting Clemson's strategic priority to deliver the No. 1 Student Experience. This will be evidenced by abstract thinking, active experimentation, application, and reflection.*

In this course, students will experience the following:

Concrete Experience: *doing/engagement*

A student shadows a physician during patient rounds, observing the physician's interaction with patients, diagnosis discussions, and treatment decisions in real time. The student is involved in the experience by watching and, when appropriate, asking questions to understand the physician's approach.

Reflective Observation: *analyzing/description and evaluation*

After the rounds, the student writes in a journal that captures a discussion with the physician about what they observed, focusing on specific patient interactions and decisions. They reflect on questions like: What techniques did the physician use to put the patient at ease? How were complex medical terms explained? This reflection allows the student to process what they saw and begin connecting it to their own thoughts and prior knowledge.

Abstract Conceptualization: thinking/interpretation

The student considers how the observed interactions and medical decisions fit within broader medical knowledge and patient care principles. They might analyze how the physician's bedside manner impacted patient trust or how the diagnosis process aligned with what they have learned about clinical reasoning. They link these observations to theoretical frameworks in patient communication and diagnostic methodology.

Active Experimentation: *testing/problem solving*

In a simulated setting or through a role-playing exercise, the student practices patient communication and diagnostic questioning, drawing on the techniques observed during shadowing. By applying the insights gained, they adapt their approach, experimenting with diverse ways to explain a diagnosis or interact empathetically with patients. This application reinforces learning and builds confidence for future clinical interactions.

AI Policy

This component should contain information regarding the classroom approach to AI usage. For recommendations on possible language, please consult the AI Use Statements documentation.

CECAS Resources Geared for STEM Majors

PEER&WISE offers free tutoring and academic support to all students in the College of Engineering, Computing, and Applied Sciences (CECAS), across all departments in Engineering and Computing. Our tutoring program focuses on high-demand and historically challenging courses, and all tutors are CRLA Level 1 certified to ensure high-quality academic support. Whether you're preparing for exams, reviewing complex concepts, or improving your study strategies, our trained team is here to help you succeed.

To learn more and access tutoring resources, visit: https://www.clemson.edu/cecas/departments/peer-wise/students/resources/

For additional information or questions please email peerwisetutoring@clemson.edu or to schedule an appointment please email for scheduling information via CU Navigate.

Spring 2026 Academic Calendar

January

- Wednesday, January 7: Classes Begin
- Tuesday, January 13: Last day to register or add a class or declare Audit
- Monday, January 19: Martin Luther King Jr. holiday (University closed, classes do not meet)
- Wednesday, January 21: Last day to drop a class or withdraw from the University without a W grade
- Wednesday, January 28: Last day to apply for May graduation

February

- Friday, February 27: Last day to issue midterm evaluations

March

- Friday, March 13: Last day to drop a class or withdraw from the University without final grades
- Mon-Fri, March 16-20: Spring break (University Open, classes do not meet)

April

- Monday, April 6: Registration for summer and fall terms begins
- Thurs-Fri, April 23-24: Classes meet; exams permitted in labs and one-hour courses only
- Monday, April 27 - Friday May 1: Examinations

May

- Monday, May 4: 9:00 AM - Deadline to submit candidate grades
- Wednesday, May 6: 9:00 AM - Deadline to submit other grades

*Academic Calendar dates are subject to change as conditions warrant.

University Policies

You can find information about campus wide policies, including student affairs information and accessibility services, by clicking on [Clemson's University Policies Page]. There, you can find information about academic integrity, access and equity (including student accessibility and Title IX info), student financial services, emergency planning, and more. My expectation is that you will review these policies carefully and be responsible for them this semester.

Student Accessibility Services

Clemson University values the diversity of our student body as a strength and a critical component of our dynamic community. Students with disabilities or temporary injuries/conditions may require accommodations due to barriers in the structure of facilities, course design, technology used for curricular purposes, or other campus resources. Students who experience a barrier to full access to this class should let the instructor know and make an appointment to meet with a staff member in Student Accessibility Services as soon as possible. You can make an appointment by calling 864-656-6848, by emailing CUSAS@clemson.edu, or by visiting Suite 239 in the Academic Success Center building. Appointments are strongly encouraged – drop-ins will be seen, if at all possible, but there could be a significant wait due to scheduled appointments. Students who have accommodations are strongly encouraged to request, obtain, and send these to their instructors via SAS as early in the semester as possible so that accommodations can be made in a timely manner. It is the student's responsibility to follow this process each semester.

Title IX

The Clemson University Title IX statement: Clemson University is committed to creating and continuously fostering a caring community based on the core values of integrity, honesty and respect. Sexual discrimination, which includes sexual harassment, sexual violence, stalking and domestic and/or relationship violence, is unacceptable and has no place in Clemson's community. Consistent with its Title IX obligation, the University prohibits discrimination, including sexual and gender-based harassment and violence, in all its programs and activities, including academics, employment, athletics, and other extracurricular activities. This Title IX policy is available online. Katherine Weathers is the Clemson University Title IX Coordinator and VP of Inclusive Excellence. She can be reached at (864) 656-3413 or via email at kweath3@clemson.edu. Remember, email is not a fully secured method of communication and should not be used to discuss Title IX issues.

Academic Integrity

As members of the Clemson University community, we have inherited Thomas Green Clemson's vision of this institution as a "high seminary of learning." Fundamental to this vision is a mutual commitment to truthfulness, honor, and responsibility, without which we cannot earn the trust and respect of others. Furthermore, we recognize that academic dishonesty detracts from the value of a Clemson degree. Therefore, we shall not tolerate lying, cheating, or stealing in any form. Using materials generated using artificial intelligence (AI) that are turned in without attribution is considered plagiarism.

All infractions of academic dishonesty by undergraduates must be reported to Undergraduate Studies for resolution through that office. In cases of plagiarism instructors may use the Plagiarism Resolution Form. See the following resources:

- Undergraduate Academic Integrity Policy
- Graduate Student Handbook

Online Testing

By enrolling in a course with online assessments, the student agrees and consents to the use of an online test proctoring service or software program as described above. The student also agrees to allowing the student, their activity, and surrounding workspace to be recorded by video and audio and then analyzed by the test proctoring system, the course instructor, and others at Clemson University. For more information see the Academic Regulations section of the Undergraduate Catalog.

Tech Support

If you have trouble with Canvas or another university system, check here first: Clemson System Status. There, you can see if there is a current issue and when it might be resolved. If there's no current issue listed, try logging out and quitting your browser before trying again. You can also look in the Canvas Help Guides for more information.

The CCIT Support Center at Clemson offers a wide range of support options and hardware repair with several contact methods to help you answer your questions as quickly as possible:

- Phone: (864) 656-3494

- Email: ITHelp@clemson.edu
- Entire catalog of IT services, including an online chat and the Knowledge Base at: IT Help and Support

Officially, Canvas supports all the major web browsers: Chrome, Firefox, Edge, and Safari. However, Safari is not fully compatible; images often do not show up in Safari, so Clemson Online recommends against using it with Canvas.

Canvas also has mobile apps available for teachers, students, and parents, all of which can be found in your mobile device's app store. These apps are convenient, but do not necessarily display or connect to every component of a Canvas site. They are good for assignment reminders, announcements, and messaging, but not as good for reading materials, taking exams, or building elements.

Emergency Preparedness Statement

Emergency procedures can be found on the Clemson University Emergency Management website and in the purple-covered flipbooks located in offices and classrooms through campus. Students and employees should ensure they are signed up to receive CU Alert text messages, familiarize themselves with emergency procedures and follow all directions from emergency responders in the event of an emergency. More information about safety resources available at Clemson University can be found at Clemson.edu/CUSafety.

Clemson Libraries

Do you need academic sources but don't know where to start?

Don't spend hours searching on your own -- ask a librarian! The Libraries provide time saving online guides for subject areas that can help you find articles, databases, books and more. Personalized research assistance with librarians who specialize in subject areas is available by appointment. Help is available in person at each of our many locations and you can also chat with a librarian live from our website, or text 864.762.4884.

Do you need to verify a book, article, or data set suggested by AI to support your research or paper?

You can check out the online catalog, use the Libraries AI Research Assistant, use verified AI research tools embedded in databases like JSTOR and EBSCO, or chat with a librarian for quick help. There is also an online guide about AI Tools for Research and a 10 Step AI Challenge that you can explore to find out more about AI tools.

Not sure how to cite a source properly in your bibliography?
There are online guides to help you with any citation style as well as citation management tools available for Clemson students: RefWorks & Endnote. There are also interactive online courses with tips to help you understand and avoid plagiarism.

Working on a creative project like creating a video or 3D object?
For assistance with digital and hands-on creative projects, the Adobe Studio and Makerspace is located on the 5th floor of Cooper Library and is staffed with experts who can help turn your creative ideas into reality. You can even check out many different kinds of equipment like high resolution cameras, green screens, and projectors from our library services desk in Cooper Library.

Need help making a chart or graph to help visualize your data?
The Data Visualization Lab on the 4th floor of Cooper provides support for data visualization, data analysis, and digital research methods.

No matter what your research needs, check out what the Clemson Libraries have to offer! www.libraries.clemson.edu

Online Course Etiquette ("Netiquette")

Even if you have experience communicating in an online environment, online courses may require an extra effort to be civil. An online classroom is still a classroom and we should all behave professionally while in it. Please read and follow the guidelines below.

While in our class, please remember:

- Be respectful. Consider the person on the receiving end of your comments. If you were sitting in front of that person, would you say the same thing to their face?
- Humor and sarcasm don’t always translate well in online discussions. Always consider your tone and whether your words can be clearly understood by your classmates.
- Only reveal information that you are comfortable sharing with your classmates.
- Please spell-check your responses; you can even compose your work in a different program (such as Word), and then paste it into the online discussion.

- Posts don't have to be perfect, but they should be as clear and readable as you can make them.
- The law still applies on the Internet. Do not commit illegal acts online, such as libeling or slandering others, and do not joke about committing illegal acts.

Please do not:

- Use offensive language, even in jest. Any comments that can be construed as racist, sexist, etc., may be removed.
- Avoid using caps lock to post messages or responses to posts. All-caps is considered SHOUTING AND IS VERY RUDE.
- Post large, unbroken blocks of text. These can be difficult to read. Try to break up your post into paragraphs.
- Post a message more than once; if you are having trouble seeing or submitting posts, contact me and we will try to fix it.
- Re-post or forward any communication that is sent to you unless you receive explicit permission to do so. Respect the response you get if your request is denied.

Refereneces

- Ian H. Witten, Eibe Frank, Mark A. Hall, Christopher J. Pal. Data Mining: Practical Machine Learning Tools and Techniques (Morgan Kaufmann Series in Data Management Systems), 4th Edition, ISBN: 978-0141988450
- http://www.kdnuggets.com/
- http://www.kdd.org/
- https://ieeexplore.ieee.org/xpl/conhome/1000179/all-proceedings